

Orange(Yingzhi) Ao

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Education

Nanjing University

2021-2025

B.A., Financial Engineering

Ranking: 3/31 | GPA: 92/100

Columbia University

2024

Exchange Student

GPA: 4.17/4.33

Relevant Courses

CS And Stats

**DS in Finance And Insurance [A+], **Predictive Modeling (in R), Statistical Machine Learning, Time Series, *Financial Engineering, *Artificial Intelligence, *Data Mining, *Options Investing Algorithms

Mathematics

**Stochastic Processes [A+], Numerical Analysis (in MATLAB), Discrete Mathematics [rank 1/23], ODE, *PDE, *Financial Mathematics

* is added before courses in progress;

** is added before grad-level courses.

Research Experiences

Multi-Period Compliance MFG-FBSDEs with DL

RA

2024.05 - Present

Supervisor: Steven Campbell

Columbia University

- **Innovation.** To solve **multi-period MFG** model in REC markets, I renovated **Deep FBSDE Solvers** and integrated numeric tricks for greater stability.
- **Results.** Presented by Prof. Campbell in the 2024 INFORMS Meeting. It will be made more realistic by adding more periods and/or inventory carryforward rules. Please check out [the Report](#) and [my GitHub repository](#) for details.

Insurance Pricing And Risk Modeling with ML

RA

2024.05 - 2024.08

Supervisor: Carlos Arocha

Arocha Association, Columbia University

- **Innovation.** Trained **K-Means** to identify risk clusters and customized **exposure rating** models for each cluster to price the insurance policies, which have not been implemented by the industry.
- **Results.** Regularly delivered Statements of Work. Presented the strategy and ML techniques in a professional manner accessible to both technical and non-technical clients.

Low-carbon Transition of China's Power System

RA

2023.02 - 2023.08

Supervisor: Yang Qu

Tsinghua-CTG Joint Center

- **Co-Authorship.** Co-authored the research report series "Low-Carbon Transition of China's Power System" and assisted in drafting and reviewing to uphold report quality.
- **Data Support.** Visualized data by plots/tables throughout the series to enhance statistical insights.

Project Experiences

Pairs Trading Optimization with GA

Independent Researcher

2023.08 - 2023.09

Applied **Genetic Algorithm** to optimize the spread weights in a cointegrated portfolio. Developed an enhanced Bollinger strategy with double bounds to improve timing for trading the optimized spread.

WorldQuant Brain

Consultant

2023.09 - 2023.12

Secured a gold medal (11th place out of 200+ participants) in the 2023 Brain x NJU Alphathon. Also contracted as a consultant to provide alpha ideas and received allowances.

Internship Experiences

Higgs Assets

High-Freq Quant Researcher

2024.08 - Present

Developed a peer-grouping system using **Affinity Propagation** combined with **XGBoost**, which will be applied to boost the Multi-Head Attention Mechanism in the next phase.

TrexQuant Investment

Global Alpha Researcher

2024.06 - 2024.08

Developed alphas based on studies in behavioral finance. Applied regression and clustering techniques.

Chongwen Quantitative

Quant Strategy Researcher

2023.09 - 2024.02

Adapted and optimized the Whitney George Strategy to the CHN stock market. Built a trading simulation to generate instructions for traders. It's been **actively used for trading** since 2024 Q2.

Skills

Programming
Deep Learning with PyTorch, markdown, L^AT_EX Git commands. Basic R, C++, and MATLAB.

English Proficiency
111/120 in TOEFL and 333/340 in GRE, with overseas experience.

Writing and Typesetting
Academic and professional reports, statements of work, and presentations.

Collaboration And Communication
Skilled in providing constructive feedback and adaptable in a team through teamwork experience.